Demo

Doing logistic regression

Interpreting Logistic Regression

Let's say this is our logistic regression model for testing the association between systolic blood pressure and diastolic blood pressure:

```
Call:
glm(formula = depress_dx ~ bp_dx, family = binomial,
data = hrs_clean)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.8163     0.1112 -16.331  < 2e-16 ***
bp_dx         0.5635     0.1257   4.482 7.39e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2348.6  on 2354  degrees of freedom
Residual deviance: 2327.0  on 2353  degrees of freedom
AIC: 2331
```

Outcome (Dependent Variable)

Predictor (Independent Variable)

```
Call:
glm(formula = depress_dx ~ bp_dx, family = binomial,
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Coefficients:
            Estimate Std. Error z value Pr(>|z|)
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    Null deviance: 2348.6  on 2354  degrees of freedom
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```

Having a hypertension diagnosis is associated with a 0.5597 increase in the log-odds of depression diagnosis compared to not having hypertension.

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.8163     0.1112  -16.33  < 2e-16 ***
bp_dx         0.5597     0.1258    4.45 8.61e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’
```

Null hypothesis: The coefficient is zero (no association).
Alternative hypothesis: The coefficient is not zero (an association exists).

We typically report odds ratios (ORs), calculated by exponentiating regression coefficients.

E.g., OR = 1.75: Individuals with hypertension have 1.75 times the odds of a depression diagnosis compared to those without hypertension

Knowledge Check

True or False: According to the model, older age is associated with lower odds of depression diagnosis, after accounting for hypertension and sex.

```
Call:  glm(formula = depress_dx ~ bp_dx + age_2014 + female,
family = binomial, data = hrs_clean)

Coefficients:
(Intercept)         bp_dx      age_2014        female
    1.01176       0.64775      -0.04605       0.86742

Degrees of Freedom: 2354 Total (i.e. Null);  2351 Residual
Null Deviance:       2349
Residual Deviance: 2230    AIC: 2238
```

a) True

b) False



University of California
San Francisco